Teaching guide: Designing Population Health Studies

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Plan and timing

The deck runs about 60 minutes of teaching, leaving roughly 15 minutes of a 75-minute class for questions and overruns. Each slide below lists what to **say**, what to **ask**, what to **watch for**, and the **transition** to the next slide. Lines in quotation marks are suggested wording, not a script to read.

Part	Slides	Minutes
1. Framing	1–4	9
2. The designs	5–10	21
3. Measuring association	11–12	10
4. Getting it right	13–16	13
5. Beyond the numbers	17–20	7 + questions
Total		**60**

Before class

- [ ] Render the deck on a machine with internet access, so the equations display
- [ ] Press **S** in the browser to open speaker view; the notes for each slide appear there
- [ ] Ask students to have a calculator or phone ready for slide 12
- [ ] Check that the arrow key reveals answers one at a time on slides 10 and 12

Part 1 — Framing (slides 1–4, 9 min)

Slide 1 — Title · 1 min

Say: "So far we have described health: who gets sick, and where. Today we ask how we *know* that something causes disease. The answer depends on how the study was built."

Transition: "Here is what you should be able to do by the end of class."

Slide 2 — Today · 1 min

Say: Read the four objectives quickly, then point to the box. "Every study asks the same question: is an exposure linked to a health outcome? What changes is where the investigator starts. Keep that sentence in mind all class."

Slide 3 — The E–D framework · 3 min

Say: "We need two letters. *E* is the exposure: something people have or do, like smoking, or something we give them, like a vaccine. *D* is the disease, or any health outcome. A 1 means present, a 0 means absent."

Point to the table: "Every study fills this 2×2 table. Cell *a*: exposed and sick. Cell *d*: unexposed and healthy."

Ask: "If *E* is texting while driving, what could *D* be?" (crashes, injuries, deaths)

Watch for: students reading *D* as disease only. It can be an injury, a death, or any measurable health outcome.

Transition: "So how do we decide which kind of study to run?"

Slide 4 — Classifying study designs · 4 min

Say: Walk the tree with one question: "Does air pollution cause asthma?"

1. "Can we assign air pollution to people? No, that would be unethical. So the study is observational."
2. "Do we compare groups? Yes, so it is analytical."
3. "Now, where do we start? With exposure, then follow forward: a cohort. With asthma cases, then look back: case-control. Both at once: cross-sectional."

"The grey boxes exist, but we won't cover them today."

Transition: "Let's take each design in turn, starting with the simplest."

Part 2 — The designs (slides 5–10, 21 min)

Slide 5 — Cross-sectional studies · 3 min

Say: "A cross-sectional study is a snapshot. We survey a sample once and record exposure and disease at the same moment. National surveys like NHANES work this way." "It is fast and cheap, and good for measuring how common something is: prevalence. The big problem is that we can't tell what came first."

Ask: "A survey finds that people who sleep badly are more often depressed. Does poor sleep cause depression, or the reverse?" (We cannot tell from a snapshot.)

Watch for: the "long-lasting disease" limitation. People who die quickly or recover quickly are missing from the snapshot, so long illnesses are over-represented.

Transition: "What if the disease is rare? Then we start from the cases."

Slide 6 — Case-control studies · 4 min

Say: "We start with people who already have the disease, the cases, and a similar group without it, the controls. Then we look back: were they exposed?" Point to the backward arrows. "This is how Doll and Hill showed in 1950 that lung cancer patients were more likely to have smoked."

"It is efficient: for a rare cancer, you don't need to follow 100,000 people for 20 years."

Watch for: recall bias. "If your child has leukemia, you search your memory much harder for past exposures than the parent of a healthy child does."

Say also: "Because we chose how many cases and controls to include, we can't compute risk. We use an odds ratio instead; more on that in a few minutes."

Transition: "Now flip the direction: start from the exposure."

Slide 7 — Cohort studies · 4 min

Say: "A cohort starts with exposed and unexposed people, all healthy at the start. We follow them forward and count who gets sick. Because we count new cases, we get real risk and the relative risk."

Point to the timeline: "Prospective and retrospective cohorts follow the same logic. The only difference is when the researcher arrives: at the start, or years later, working from records." Example: the Framingham Heart Study, running since 1948.

Ask: "Why is a cohort a poor choice for a disease that affects 1 person in 100,000?" (You would need an enormous cohort to see even a few cases.)

Transition: "In all three designs, we only watch. What if we decide who gets exposed?"

Slide 8 — Randomized controlled trials · 4 min

Say: "In a trial, the researcher assigns the exposure by chance, like a coin flip. Randomization makes the two groups alike on everything, even things we never measured. That is why the RCT is the strongest design for cause and effect."

Optional story (1 min): observational studies suggested that hormone replacement therapy protected women's hearts. The Women's Health Initiative trial (2002) found the opposite for estrogen plus progestin. Women who chose the therapy were healthier to begin with; randomization removed that difference.

Ask: "Why can't we run a trial on smoking?" (Assigning a known harm is unethical.)

Mention the drug-trial phases in one sentence (footnote on the slide).

Transition: "Let's put the designs side by side."

Slide 9 — Designs at a glance · 1 min

Say: "Don't memorize this table; read it by row. Designs differ on who chooses the exposure, what we select people on, and which way time runs. This is a good page for your cheat sheet." Read the box.

Transition: "Let's test it."

Slide 10 — Which design is it? · 5 min

Do: read each scenario, give about 30 seconds, take a show of hands, then press the arrow key to reveal the answer.

1. **Case-control** — we start from children who already have leukemia, a rare disease.
2. **Cross-sectional** — one survey, exposure and outcome at the same time.
3. **Prospective cohort** — exposure recorded first, then followed forward.
4. **Cluster RCT** — schools, not individual students, are randomized.

Watch for: students calling scenario 3 an RCT. Nobody assigned smoking; the nurses chose it.

Part 3 — Measuring association (slides 11–12, 10 min)

Slide 11 — Measuring association: RR and OR · 4 min

Say: "Now we put a number on the association. Relative risk is the risk in the exposed divided by the risk in the unexposed: *a* over *a* + *b*, compared with *c* over *c* + *d*. An RR of 2 means the exposed are twice as likely to get the disease."

"In a case-control study, we chose how many cases and controls to include, so 'risk' has no meaning. We use the odds ratio instead: *a* times *d* over *b* times *c*."

Point to the box: "Read both the same way. 1 means no association; above 1, higher risk; below 1, lower risk, which is protective."

Ask: "A vaccine trial reports an RR of 0.5. Good news or bad news?" (Good: the vaccinated have half the risk.)

Transition: "Let's compute both on the same data."

Slide 12 — Worked example · 6 min

Do: before revealing anything, give 2 minutes to compute RR and OR for the left-hand table. Then press the arrow key.

Say: "Risk is 3% against 1%, so the RR is 3.0. The OR is 3.1: almost the same."

Ask, before the second reveal: "When the outcome is common, will the OR and the RR be closer or further apart?" Then reveal: RR 2.0, OR 2.7.

Say: "Rule of thumb: when the outcome is rare, OR ≈ RR. When it is common, the OR exaggerates the association."

Watch for: students computing a/b instead of $a/(a + b)$. Risk always uses the row total.

Part 4 — Getting it right (slides 13–16, 13 min)

Slide 13 — Error, reliability, and validity · 3 min

Say: "Every study makes errors, of two kinds." Point to the top-right target: "Random error scatters the shots. That is low reliability. More data helps, because the errors average out." Point to the bottom-left target: "Systematic error pushes every shot the same way. That is bias, low validity. More data does not help: you just get a very precise wrong answer."

"The bottom-left target is the dangerous one: consistent, confident, and wrong."

Point to the box: "A study of male factory workers may be right for them, which is internal validity, and still not apply to women or children, which is external validity."

Transition: "So where does bias come from?"

Slide 14 — Bias: two main families · 3 min

Say: "Two families: who gets into the study, and how we measure them."

On the *Selection bias* tab: "Compare factory workers with the general population and work looks healthy. But the sickest people could not work in the first place."

Click the *Information bias* tab: link recall bias back to slide 6. "Ask students how much they drink, and the answer comes in low. That is social desirability bias."

Ask: "A survey on exercise habits is handed out at a gym. Which bias?" (selection, or volunteer bias)

Transition: "There is a third problem, and it is not a measurement mistake at all."

Slide 15 — Confounding · 4 min

Do: before explaining anything, ask: "Coffee drinkers get more lung cancer. Does coffee cause lung cancer?" Let students find smoking themselves.

Say: "Smoking is a confounder. It travels with coffee, and it causes lung cancer, so it creates an association that is not causal." Read the three conditions. "We control it by design: randomize, restrict the study to non-smokers, or match on smoking. Or by analysis: compare within smokers and within non-smokers, or use regression."

Watch for: the third condition. If coffee caused lung cancer *through* some step, that step would be a mediator, not a confounder. One sentence is enough; mediators are new to them.

Transition: "Sometimes a third variable does not distort an effect. It changes it."

Slide 16 — Effect modification · 3 min

Say: "Look at the table. Asbestos alone: about 5 times the risk. Smoking alone: about 11 times. Both: 53 times. If the effects simply added up, we would expect about 15. Together they are far worse than the sum: that is synergy."

"This is not a bias to fix; it is a finding. Asbestos workers who smoke are at extreme risk, so they are the first people to help quit."

Watch for: confusion with confounding. "Confounding is a distortion we remove. Effect modification is real, and we report it." The additive and multiplicative formulas are in the speaker notes; keep them off the board.

Part 5 — Beyond the numbers (slides 17–20, 7 min + questions)

Slide 17 — Qualitative and mixed methods · 2 min

Say: "Numbers tell us how much; qualitative methods tell us why." Name the four techniques in one breath. "They help before a survey, to write good questions; during a study, to explain surprises; and after it, to make sense of the results."

Point to the example box: "Your hometown community assessment paper uses exactly this mix of data and people's own accounts."

Slide 18 — Ethics in population health research · 3 min

Say: read the four principles quickly. "These rules exist because research has done real harm."

Story (1 min): in the US Public Health Service study at Tuskegee (1932–1972), Black men with syphilis were left untreated for decades, even after penicillin became the standard cure. Public outrage led to the National Research Act (1974), which formalized IRBs, and to the Belmont Report (1979).

Ask: "Which of the four principles did Tuskegee violate?" (All of them. No informed consent violates autonomy, and putting the burden on poor Black men violates justice.)

Slide 19 — Key takeaways · 2 min

Say: read the five points, then the box. "If you remember one thing: where the investigator starts defines the design."

Slide 20 — Questions

Ask: an exit question, if no one has questions: "Pick a health question you care about. Which design would you use, and why?" Take two or three answers.

If you run late

These cuts save about 8 minutes. Keep slides 6, 11, 12 and 15 intact: they carry the calculations and the main concepts.

- **Slide 8:** drop the hormone replacement therapy story (−1 min)
- **Slide 9:** skip it and point students to it for their cheat sheet (−1 min)
- **Slide 10:** do two scenarios instead of four, for example 1 and 3 (−2 min)
- **Slide 14:** show one tab and name the other family in one sentence (−1 min)
- **Slide 17:** one sentence plus the example box (−1 min)
- **Slide 18:** shorten the Tuskegee story to one sentence (−1 min)
- **Slide 19:** read only the box (−1 min)